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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

MINI PROJECT SYNOPSIS

Subject Name	Mini Project Work	Subject Code	BCSP586
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Domain	WEB APPLICATION DEVELOPMENT HEALTHCARE INFORMATICS ARTIFICIAL INTELLIGENCE IN HEALTHCARE	Group No:	A19
Project Title	AI-Driven Women's Health Web App for PCOS/PCOD and Maternal Care		
Guide Name	Kavitha B		

ABSTRACT:

This project focuses on creating an interactive web application dedicated to women's health, with a special emphasis on PCOS/PCOD management and maternal care. I chose this topic because these issues are common but often not addressed openly, leaving many women without proper guidance or support. The idea is to build a platform that combines awareness, self-tracking tools, and professional advice in a way that is easy to access and culturally relevant.

The application is designed to provide AI-based diet and workout recommendations tailored to each user's profile. It allows users to track their menstrual cycles, symptoms, pregnancy milestones, and lifestyle habits through an intuitive and responsive interface. A community forum is included to encourage peer support and experience sharing, while a secure back-end ensures the privacy of sensitive health data. The platform also connects users with verified doctors, nutritionists, and mental health professionals for more personalized help.

By integrating educational resources, tracking features, and expert guidance into one system, this project aims to empower women to take better control of their health. Ultimately, I want it to be a reliable companion that bridges the gap between awareness and action, helping women make informed decisions for their well-being.

OBJECTIVES:

- To build a user-friendly web application for women's health awareness and self-care.
- To provide AI-driven diet and workout plans tailored for PCOS/PCOD management.
- To enable menstrual cycle and pregnancy milestone tracking through an interactive interface.
- To create a secure platform for storing and managing personal health data.
- To connect users with verified healthcare professionals for expert guidance.
- To foster a supportive community where women can share experiences and advice.
- To offer easy-to-understand educational resources on PCOS/PCOD and maternal health.

INTRODUCTION ABOUT THE DOMAIN:

The project brings together multiple domains to create a comprehensive women's health platform. The primary domain is **Healthcare Informatics**, which focuses on using technology to collect, process, and present health-related information in an accessible way. This allows the application to store and manage personal health data securely while offering AI-based insights for better lifestyle and medical decisions.

Another important domain is **Web Application Development**, which is the backbone of the project. It involves designing and building an interactive, responsive, and easy-to-use interface that works seamlessly across devices. This ensures that users can access the platform anytime, anywhere, whether to check their cycle, log symptoms, or connect with experts.

The project also draws from the domain of **Artificial Intelligence in Healthcare**, specifically in personalizing diet and workout recommendations for PCOS/PCOD. By using AI models, the system can adapt to each user's health profile, preferences, and progress over time.

Lastly, the **Database Management Systems (DBMS)** domain ensures structured and secure storage of user health logs, forum posts, and other important records. Together, these domains form the technological and functional foundation of the project, enabling it to be both informative and user-centric.

REQUIREMENT SPECIFICATION:

Minimum Hardware Requirements

- **Processor:** Dual-core (Intel i3 7th Gen or AMD equivalent)
- **RAM:** 4 GB
- **Storage:** 120 GB SSD or 250 GB HDD

- Display:** 1366×768 resolution
- Internet:** Stable connection (≥ 2 Mbps)
- OS:** Windows 10, macOS, or Linux

Minimum Software Requirements

Frontend(HTML,CSS,JS,CALENDAR,REMINDER-DASHBOARD AND BABYGROWTH TRACKER)
Backend(PYTHON,FLASK,DJANGO,HEALTH RECORD DATABASE AND REMINDER SCHEDULE)

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Signature of Guide

Mini Project Coordinator